



DIY electronics for everyone

From drones, to robotics to home automation, DIY electronics are no longer only for nerds.
<https://digit.in/sep18-21-1>



Bizarre Chine Firewall bans

Three strange things banned by the Great Firewall of China include Peppa Pig and the letter N.
<https://digit.in/sep18-21-2>

cal computers actually have several advantages over electronic computers - they are resilient to electric shocks, can operate at much higher temperatures, and are more energy efficient.

There are some extreme environments that are just hostile to semiconductor based computers. One of these environments is the surface of Venus. Venus has an atmospheric pressure of 90 atm, which is 90 times that of the Earth. The amount of pressure is enough to crush most submarines. The surface temperature 462°C, hot enough to melt lead alloys. This is where the reliable and old school mechanical computers come to the rescue. NASA is planning to send a clockwork rover to Venus. The rover would navigate using tank like treads, powered by turbines within the body of the rover. The rover can use Morse code for communicating with a satellite in orbit. A rotating sheath over a bright radar target can repeatedly cover and uncover the target to send signals to the orbiting satellite. Engineers working on the rover are figuring out a way to integrate a drill as well.

REDSTONE COMPUTERS

Redstone dust is a kind of material in Minecraft that can be used to make virtual computers. Redstone can be laid out in lines to make wires. Players can also craft redstone clocks and rails. Typical computers built within Minecraft consist of redstone, redstone torches and redstone repeaters. The redstone torches have an on or off state, and the repeaters can amplify, delay or block signals. Redstone pistons are used for the on and off state of binary. Combined with a number of buttons and levers, elaborate computing devices can be made within the game. The amount of time spent into creating computers is staggering, and includes ALU, CPU, memory, input and output devices.

Elaborate community creations included a scientific calculator, a word processor, and even a working version of the Commodore 32. Users went on to create programmable comput-

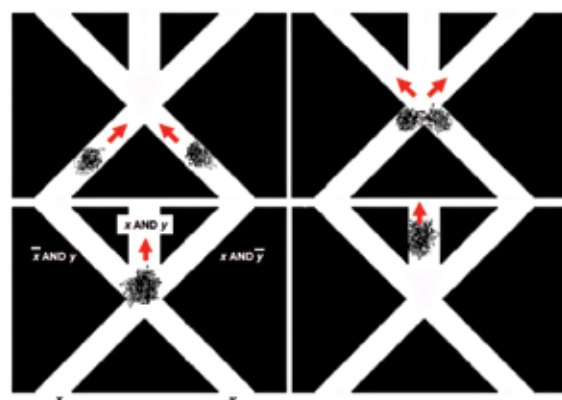
ers, and a hard disk drive that could actually store 1KB of memory. Many of the more elaborate creations included MP3 players and had their own AI for games such as tic-tac-toe. Hans Lemurson made a 2D version of Minecraft, within Minecraft. Players can move around, and create or destroy blocks. A 2012 update introduced command blocks, that interacted with redstone, and allowed the computers to be significantly miniaturised. This enabled another player, Crafternics, to create a working version of the iPhone. It had a keyboard, a working weather app, and you actually had to enter a passcode to access the device. The build used 2572 command blocks, and over 200 hours to create. Computing is a terribly slow process within the virtual Minecraft computers, though there are plugins to accelerate how fast the game plays out.

BIOLOGICAL COMPUTERS

Chapter 17, titled Newton, Von Neumann, the First Emperor, and Tri-Solar Syzygy in Cixin Liu's award winning sci-fi book, The Three Body

Problem has a spectacular description of a computer. The basic units are soldiers who perform simple operations, raising black or white flags to form AND, XOR, NOR, XNAND and XNOR gates. The hardware consists of thirty million soldiers, arranged in an area of six square kilometers. The NOT gate is simply a soldier who raises the opposite coloured flag, when another soldier raises a particular flag. There is a CPU, registers, memory, a hard drive and even an expandable bus system. The software is an operating system on a scroll. There is even a display unit with a progress bar made up of a series of green flags. Even if it didn't work out in the story, it is a brilliant demonstration of how complex machines can be built out of very simple components - like people themselves.

Researchers at Japan's Kobe university used crabs to build a computer. They used soldier crabs from Australia, which behave very predictably in swarms. The gates, or the most basic components of the computer, were mazes through which the crabs were made to navigate. Depending on how many swarms were introduced into the legs, the crab swarms would move consistently through the maze. An inverted Y shaped maze was used as an OR gate, and an X shaped maze with an extra arm on one side was used as an AND gate. The remaining gates could be made by combining these two gates together. No crabs were harmed during the experiment, and the researchers went to the extent of using



AND Gate implementation using crabs

Credit: Yukio-Pegio Gunji, Yuta Nishiyama and Andrew Adamatzky

cork for building the maze so that it was "comfortable for crab's friction".

German and British researchers have created logic components from slime mould. The species, known as Physarum polycephalum, forms networks of tubes to absorb and transport nutrients. The mould aggregates around food particles, but avoids salt. By combining the salt and oat flakes, structures similar to circuits seen in computers could be formed. Using dyes made up of magnetic nanoparticles, XOR and NOR gates were constructed. A series of these gates can potentially be chained up to build complex computers. A practical application of a system like this is that the mould can help humans build more robust computer and mobile communication networks. 📌